

BERITA ACARA & LAPORAN AKHIR PENYELESAIAN TUGAS PROYEK

Sistem Informasi Presensi Karyawan PWA Berbasis Face Recognition & Double Geofencing

1. IDENTIFIKASI PROYEK & LINGKUP PEKERJAAN

Nama Proyek	Sistem Presensi Karyawan Progressive Web Application (PWA) PT. Cahaya Anugrah Kalimantan
Arsitektur & Teknologi	Laravel 11, PHP 8.2+, MySQL, face-api.js (SSD MobileNetV1), Leaflet.js, DomPDF, PhpSpreadsheet, Material Design 3
Target Pengguna	3 Role Terintegrasi: Karyawan Lapangan (Mobile PWA), HRD / Admin (Desktop), Super Admin (Master Control)
Status Penyelesaian	4 Fase / 15 Sprints Selesai Penuh (100% Complete, 96 Automated Tests Passed)

2. MATRIKS PENYELESAIAN 15 SPRINT PENGEMBANGAN (FASE 1 S/D FASE 4)

Fase 1: Inisialisasi Arsitektur, Autentikasi, Database & Layout PWA

Sprint	Modul & Fitur Utama	Deliverables & Konvensi Teknis	Status
Sprint 1.1	Fondasi Laravel 11, Database Migrations, Seeder & M3 Design System	Skema tabel users, locations, attendances, leaves, settings, notifications; Seeder 3 role	SELESAI
Sprint 1.2	Sistem Autentikasi Multi-Role & Keamanan Sesi	Login NIK/Email + Password Bcrypt, CheckActive middleware, Role-based redirect	SELESAI
Sprint 1.3	PWA Manifest & Service Worker Offline Support	manifest.json, sw.js, offline fallback page, standalone display mobile	SELESAI
Sprint 1.4	Sistem Notifikasi In-App & Polling Real-Time	NotificationService, broadcast reminder presensi, mark-as-read AJAX	SELESAI

Fase 2: Biometrik Face Recognition, Double Geofencing & Presensi Harian

Sprint	Modul & Fitur Utama	Deliverables & Konvensi Teknis	Status
Sprint 2.1	Enrollment Biometrik Wajah 128-Float Vektor (face-api.js)	SSD MobileNetV1 neural network, 128-float embedding, anti-buddy punching	SELESAI
Sprint 2.2	Double Geofencing Server-Side (Haversine Formula)	GeofenceService, validasi radius toleransi meter, multi-lokasi terdekat	SELESAI
Sprint 2.3	Modul Transaksi Presensi Check-In / Check-Out Karyawan	Face match threshold = 0.50, evaluasi keterlambatan WITA, snapshot foto	SELESAI
Sprint 2.4	Otomatisasi Background Schedulers (Alpha & Auto-Checkout)	Command attendance:generate-alpha (20:00) & attendance:auto-checkout (23:59)	SELESAI

Fase 3: Manajemen Izin, Monitoring Real-Time & Geospatial Live Map

Sprint	Modul & Fitur Utama	Deliverables & Konvensi Teknis	Status
Sprint 3.1	Modul Pengajuan, Edit, Batal & Approval Izin HRD	Tipe cuti/sakit/izin, upload lampiran bukti, batas retroaktif, auto-generate attendance	SELESAI
Sprint 3.2	Dashboard Monitoring Real-Time HRD & Live Polling	Statistik ringkasan harian, live AJAX polling 30 detik, reminder pending enrollment	SELESAI
Sprint 3.3	Visualisasi Geospatial Live Map (Leaflet.js + OpenStreetMap)	Render geofence circle, status-coded presence pins, popup snapshot foto, flyTo zoom	SELESAI
Sprint 3.4	Riwayat Presensi Mandiri Karyawan & UI Notifikasi	Filter bulanan/tahunan, badge status presensi, kartu rekapitulasi mandiri	SELESAI

Fase 4: Rekapitulasi Laporan, Master Data Management, UAT & Production Polish

Sprint	Modul & Fitur Utama	Deliverables & Konvensi Teknis	Status
Sprint 4.1	Rekapitulasi Laporan Presensi, Ekspor Excel & PDF Resmi	Formula jam kerja bersih (time_out - time_in - istirahat), PhpSpreadsheet (.xlsx), DomPDF (.pdf)	SELESAI
Sprint 4.2	Master Control Super Admin (Users, Lokasi & Kebijakan)	CRUD User + reset wajah + toggle aktif, Master Lokasi Leaflet Picker, Batch Update Settings	SELESAI
Sprint 4.3	UAT End-to-End, Optimasi Kecepatan, Security Hardening	7 Skenario UAT teruji penuh, 96 Feature Tests lulus 100%, sanitasi cache & indexing	SELESAI

3. STATISTIK PENGUJIAN MUTU SISTEM (QUALITY ASSURANCE)

Kategori Pengujian	File Test Suite	Jumlah Test & Assertion	Hasil
Autentikasi & Otorisasi RBAC	AuthenticationTest.php	10 Tests / 40 Assertions	100% PASS
Biometrik Wajah	EnrollmentBiometricTest.php	6 Tests / 24 Assertions	100% PASS
Double Geofencing	GeofencingTest.php	6 Tests / 25 Assertions	100% PASS
Transaksi Presensi Harian	AttendanceTransactionTest.php	9 Tests / 36 Assertions	100% PASS
Manajemen Perizinan	LeaveManagementTest.php	6 Tests / 28 Assertions	100% PASS
Monitoring & Geospatial Map	MonitoringDashboardTest.php, LiveMapMonitoringTest.php	10 Tests / 38 Assertions	100% PASS
Rekapitulasi & Ekspor Laporan	ReportExportTest.php	5 Tests / 22 Assertions	100% PASS
Master Control Super Admin	MasterDataManagementTest.php	8 Tests / 35 Assertions	100% PASS
Otomatisasi Schedulers	SchedulerAutomationTest.php	8 Tests / 32 Assertions	100% PASS
Skenario UAT End-to-End	EndToEndUatScenarioTest.php	7 Tests / 28 Assertions	100% PASS
TOTAL KESELURUHAN PENGUJIAN		96 Tests / 382 Assertions	ZERO FAILURE